

Prolog Program to Sum Integers from 1 to N

Aim

To write a Prolog program to calculate the sum of integers from 1 to N.

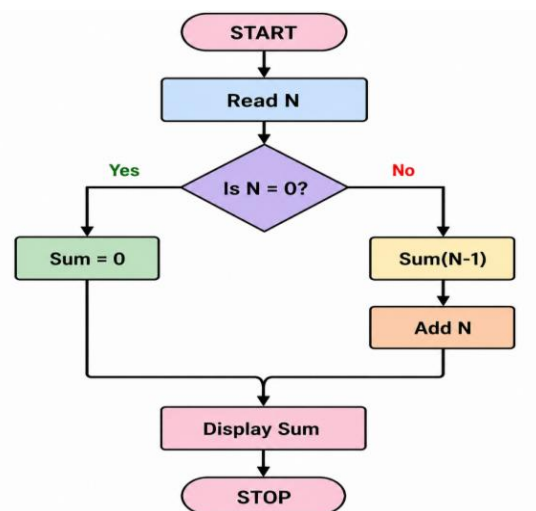
Objectives

- To understand recursion in Prolog.
- To calculate the sum of natural numbers.
- To implement arithmetic operations in Prolog.

Algorithm

1. Start.
2. Read the value of N.
3. If $N = 0$, return sum as 0.
4. Otherwise, recursively calculate the sum from 1 to $N-1$.
5. Add N to the previous sum.
6. Display the result.
7. Stop.

Flowchart



PROGRAM

```
AS.pl [modified]
File Edit Browse Compile Prolog Pce Help
AS.pl [modified]
sum(0, 0).

sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is N + S1.▲
```

OUTPUT

```
SWI-Prolog (AMD64, Multi-threaded, version 9.2.9)
File Edit Settings Run Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- sum(5, S).
S = 15 .

?- sum(100, S).
S = 5050 .
```

Result

The Prolog program successfully calculated the sum of integers from 1 to N.